

Initial one page project specification

Product – Plant nursery breeding and management system

Primary User - A single plant nursery owner

Core features

Section 1 - Plants

- Plant records
- Parentage
- Photos
- Care: watering, fertiliser, environmental conditions, soil mix
- Current location
- Growth observations / pot and plant measurements
- Status: growing, breeding, sold, deceased, etc.
- Purchase information
 - Seller
 - Purchase date
 - Price
 - Shipping / other purchase costs if needed
- Seedling / plant origin
 - Seed batch linked if grown from seed
 - Seedling / keeper status

Section 2 - Equipment

- Name
- Brand
- Model
- Purchased date
- Price
- Maintenance
- Future upgrades

Section 3 – Expenses

- Date
- kWh
- Running time per day
- Power usage / electricity cost
- Amount based on electric price
- Linked equipment
- All equipment running cost
- Other nursery expenses if needed
 - Fertiliser
 - Growing media
 - Packaging
 - Shipping
 - Other supplies

Section 4 - Breeding

- Historical breeding records
- Potential future crosses and timeline
- Mother plant
- Pollen parent
- Cross/offspring name generation
- Breeding status
- Breeding plan / potential cross
 - Priority
 - Target date
 - Notes
 - Status

Section 5 - Pollen inventory

- Source plant
- Collection date
- Storage date
- Storage method/location
- Quantity/amount if practical
- Status: available, used, expired, discarded
- Which breeding events it has been used for

Section 6 - Seed batches

- Parent cross
- Pollination date
- Infructescence development
- Fruiting
- Seed development
- Harvest
- Number of seeds
- Germination results
- Seed Inventory
- Related seedlings

Section 7 - Seedlings

- Individual plant records linked to seed batch
- Selection/keeper status
- Growth observations (Impressive Phenotype, and strength of characteristics from parents)
- Photos
- Eventual parent status

Seedlings will use the main Plant record and be linked to their seed batch/origin, rather than being completely separate from Plants.

MVP

Plant management

- Add Plant
- Edit Plant
- Archive Plant
- Photos
- Parentage
- Current location
- Status
- Purchase information

Care

- Watering
- Fertilising
- Last watered
- Last fertilised
- Watering status indicator
- Fertiliser status indicator

Equipment

- Add/edit equipment
- Purchase date/costs
- Basic status/maintenance
- Running expenses

Dashboard

- Plant count
- Plant investment
- Equipment Investment
- Total nursery investment
- Watering overview (Colour changing scale on watering levels)

Future features

Breeding

- Select mother and pollen parent
- Generate potential cross name
- Create breeding record
- Past and potential cross distinction
- Breeding plans
- Predictive Phenotype Modelling / Virtual breeding
- Potential traits based on recorded parent characteristics

Ancestry

- Family tree
- Parent/offspring relationships
- Genetic lineage visualisation
- Expandable ancestry and recorded family tree data for each plant

Seed batches

- Pollination → infructescence → fruiting → seed → germination
- Seed inventory
- Seedling generation
- Seedlings linked to seed batch

Environmental integration

- Live temperature
- Humidity
- Equipment status
- Historical environmental data (How long they have been in certain conditions)

External image/search

- Related plant images
- Parent imagery
- Breeding reference images

Sales

- Plant sold
- Buyer/order
- Sale price
- Profit/loss
- Seed sales
- Seedling sales

Non Functional Requirements

- No destruction of historical records
- Archive rather than permanently delete important plant records
- Database backups
- Image optimisation
- Responsive dashboard
- Mobile friendly
- Accessible colours
- Status should not rely only on colour
- Automated tests
- Typescript
- Data validation
- Error handling

Database Ideas

Have main plant entity with everything linking to the Plant ID, like:

Plant

- identity
- parentage
- purchase
- photos
- current location
- status
- measurements
- care events
- observations
- breeding participation
- origin / seed batch

Instead of including last water and fertiliser in the Plant ID, will instead include all data within CareEvent like:

Care events

- plantID
- eventType
- date
- notes
- amount

Last watered and last fertilised will be calculated from the most recent CareEvent rather than stored as separate main Plant fields.

Water colour indication

Each plant will have a colour coded water status, which displays a visual state on how long since it has been watered, and could do the same for fertiliser.

The target number of days should be configurable for each plant instead of having one fixed number for every plant.

Something like:

- Day 0 - Blue
- Day 1 – Blue / Green
- Day 2 – Green
- Day 3 – Yellow / Green
- Day 4 – Yellow
- Day 5 – Orange
- Day 6 - Brown

The colour can gradually change based on:

- days since watered / target dry days

Breeding Section

Could model breeding events like:

breedingEventID

- motherPlantID

- pollenPlantID
- date
- status
- notes

This allows the Cross to be generated from Mother + Pollen.

Potential future crosses should be kept as Breeding Plans, separate from actual Breeding Events.

For example:

Breeding Plan

- motherPlantID
- pollenPlantID
- priority
- targetDate
- status
- notes

Then once the cross is actually made it can become a Breeding Event.

Pollen Inventory

To keep track of pollen stores, could map Pollen on the database like:

pollenId

- sourcePlantId
- collectionDate
- storageDate
- storageLocation
- storageMethod
- quantity
- status

Future upgrade section

Section for plans for future upgrade and expansion for example:

Project - Add second tent

Estimated cost: 300

Current saved:

Target date:

Status: Not yet planned

Ancestry section

Expandable ancestry and recorded family tree data for each plant, so to be able to open a plant and see

ANT-1043

HURC × Snake Oil

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└─ HURC (Seed Parent)

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└─ HU / KoS (King of Spades)

|

└─ Red Crystal

|

└─ Snake Oil (Pollen Parent)

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└─ (Black Widow × Michelle)

|

└─ (papillilaminum × magnificum)

Observation feature

Observations could be carried out periodically, maybe every 3-6 months, allows for leaf size over time to be calculated and to look at fastest growth.

Observation

- date
- plantId
- leafLength
- leafWidth
- leafNumber
- vigourScore
- venationScore
- velvetScore
- emergentColourScore
- notes
- photo

Observations can be used to compare growth and phenotype between plants and eventually between offspring from different crosses.